

Zeiad Khafagy

+1 720-590-2826 | zeiadhkhafagy@gmail.com | Ames, IA, USA | [GitHub](#) | [LinkedIn](#)

EDUCATION

Iowa State University

M.S. in Computer Science

Ames, IA

Aug. 2026 – Present

Istanbul Bilgi University

Bachelor in Computer Engineering

Istanbul, Turkey

Oct. 2022 – June 2026

- GPA: 3.13/4.00; Second Honors List

EXPERIENCE

Research Assistant

Aug. 2026 – Present

Iowa State University – Department of Aerospace Engineering

Ames, IA

- Conduct research on data-driven control methods for Wave Energy Conversion (WEC) systems using diffusion models.
- Develop and evaluate diffusion-model-based control approaches for wave energy systems operating under stochastic and irregular ocean wave conditions.
- Apply machine learning, numerical simulation, and control methods to investigate strategies for improving wave energy capture and system performance.

Undergraduate Research Assistant

Sept. 2025 – Dec. 2025

Iowa State University - Department Of Computer Engineering

Ames, IA

- Optimized a Wave Energy Conversion (WEC) system to maximize power generation from stochastic ocean waves using Deep Reinforcement Learning.
- Implemented and trained Soft Actor-Critic (SAC) and Deep Deterministic Policy Gradient (DDPG) agents, leveraging SAC's entropy regularization to address local minima observed in DDPG models.
- Increased total energy absorption by 1.14% compared with classical Bang-Singular-Bang control under irregular sea-state simulations.

Undergraduate Research Assistant

July 2024 – Oct. 2024; Feb. 2025 – March 2025

Istanbul Technical University – PIMI Lab

Istanbul, Turkey

- Developed an automated deep learning framework using nn-UNet for coronary artery segmentation from CT angiography images.
- Engineered a transfer learning pipeline using the ASOCA and ImageCAS datasets and incorporated post-processing algorithms to refine segmentation of complex anatomical structures.
- Achieved a Dice Similarity Coefficient of 0.856 and Hausdorff Distance of 14.5, improving the segmentation Dice score by 3.07% through post-processing.

PUBLICATIONS

O. Bani Hani, **Z. Khafagy**, M. Staber, A. Gaffar, and O. Abdelkhalik, "Control of Wave Energy Converters Using Reinforcement Learning," in *Journal of Marine Science and Engineering (JMSE)*, January 2026.

Z. Khafagy and I. Oksuz, "Coronary Artery Vessel Tree Segmentation Using Transfer Learning from CT Angiography Images," in *Proc. IEEE 33rd Signal Process. Commun. Appl. Conf. (SIU)*, August 2025.

TECHNICAL SKILLS

Programming Languages: Python, Java

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, nnU-Net

Data & Scientific Computing: NumPy, Pandas, Matplotlib

VOLUNTEERING

Buddy Program Volunteer

Feb. 2024 – June 2024

Istanbul Bilgi University

Istanbul, Turkey

- Assisted incoming Erasmus exchange students in adapting to academic and daily life in Turkey.
- Helped students navigate academic registration, course selection, and university systems.
- Supported cultural and social integration through peer mentorship and student meetups.